

Raj Kumar Nelluri

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PROFESSIONAL SUMMARY

AI / Machine Learning Engineer specializing in LLM-powered applications and predictive ML systems. Built and deployed a production-style Retrieval-Augmented Generation (RAG) platform on AWS enabling document intelligence with citation-grounded responses. Experienced in Python, LangChain, vector databases, and end-to-end ML pipelines from data ingestion to deployed inference.

TECHNICAL SKILLS

Languages: Python, SQL

AI / LLM Systems: Retrieval-Augmented Generation (RAG), LangChain, OpenAI API, Vector Databases, Embeddings, Prompt Engineering

Machine Learning: Scikit-learn, XGBoost, TensorFlow, Feature Engineering, Model Evaluation

Cloud & Deployment: AWS (EC2, SageMaker, Lambda, S3), Docker, REST APIs, FastAPI, Streamlit

Data & Analytics: pandas, NumPy, Matplotlib, Tableau, Power BI

AI ENGINEERING PROJECTS

Enterprise RAG Chatbot – LLM-Powered Document Intelligence | [LiveDemo](#) | [GitHub](#) 2026

Python | *LangChain* | *OpenAI* | *ChromaDB* | *Docker* | *AWS EC2*

- Designed and deployed a Retrieval-Augmented Generation (RAG) system enabling users to upload and query PDF documents with citation-grounded responses
- Implemented document ingestion pipeline including PDF parsing, chunking (1000/200 overlap), and embedding generation using OpenAI embeddings
- Built semantic retrieval pipeline using ChromaDB vector database with Maximum Marginal Relevance (MMR) search and reranking to improve answer relevance
- Added hallucination mitigation mechanisms including similarity-threshold filtering and context-grounded prompts
- Containerized application using Docker and deployed on AWS EC2 with persistent vector storage supporting a live public demo
- Achieved ~ 2 second end-to-end response latency including retrieval and LLM generation

Customer Churn Prediction Pipeline | [GitHub](#) 2025

Python | *XGBoost* | *AWS SageMaker* | *Lambda* | *CloudWatch*

Pace University, NY

- Built end-to-end ML pipeline on AWS for customer churn prediction including S3 ingestion, Lambda ETL orchestration, and SageMaker model training
- Extracted NLP features from 7,000+ support tickets using AWS Comprehend to enhance behavioral modeling
- Deployed XGBoost model as SageMaker real-time inference endpoint and implemented automated monitoring with CloudWatch and EventBridge

Retail Sales Forecasting System | [GitHub](#) 2025

Python | *XGBoost* | *Pandas* | *AWS S3*

Pace University, NY

- Built time-series forecasting model using XGBoost to predict retail demand from 500,000+ transactions
- Engineered lag features, rolling averages, and seasonal indicators improving demand prediction accuracy

Insurance Fraud Detection System 2024

Python | *Scikit-learn* | *AWS Lambda* | *Kinesis* | *RDS*

Pace University, NY

- Built time-series forecasting model using XGBoost to predict retail demand from 500,000+ transactions
- Engineered lag features, rolling averages, and seasonal indicators improving demand prediction accuracy

ADDITIONAL PROJECTS

Crypto Price Forecasting API | *Python* , *XGBoost* , *Flask* 2023

- Compared LSTM, CNN, and XGBoost models for cryptocurrency price prediction
- Deployed XGBoost model as a Flask REST API serving 7-day price forecasts

3D Face Generation (NeRF) 2023

- Implemented Neural Radiance Fields for 3D face reconstruction using JAX and TensorFlow
- Optimized GPU memory usage by 30% enabling training on standard hardware

EDUCATION

Pace University

Master of Science in Computer Science

New York City, NY

Sep 2023 – Apr 2025

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Artificial Intelligence

Bengaluru, India

Jul 2019 – Apr 2023

CERTIFICATIONS

AWS Certified Cloud Practitioner – Amazon Web Services

2025

Adobe Journey Optimizer Foundation – Adobe

2025

Complete Python Developer – Udemy

2023